

SUBRAMANIAN M

AI Engineer | Generative AI | LLM Applications | RAG | Agentic Systems

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PROFESSIONAL SUMMARY

AI Engineer focused on building production-oriented Generative AI and LLM applications with Python, LangChain and LangGraph. Hands-on experience across **LLM agents, RAG, embeddings, vector search, tool calling and multimodal AI**. Built and operated an independent AI voice receptionist product end to end, covering conversation architecture, API integrations, PostgreSQL, Docker/Linux deployment and production debugging. TensorFlow Developer Certified with a strong software-engineering foundation in TypeScript and full-stack development.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL

Generative AI & LLM: LangChain, LangGraph, LangChain Runnables / LCEL, RAG, LLM agents, tool calling, embeddings, vector search, hybrid retrieval, reranking, multimodal AI, prompt engineering, OpenAI / Claude / Gemini APIs, Ollama

ML & Data: TensorFlow, PyTorch, Hugging Face, scikit-learn, FAISS, data preprocessing, model evaluation

Backend & Infrastructure: PostgreSQL, Docker, Linux, Hetzner Cloud, Vercel, Git, REST APIs, webhooks

Automation & Integration: n8n, Firecrawl, Google Workspace / Calendar, workflow orchestration

AI ENGINEERING EXPERIENCE & PROJECTS

LazyLoop - AI Voice Receptionist Platform | Independent Product | Vapi, Next.js, Node.js, PostgreSQL

- Built and deployed an **AI voice agent** that handles live inbound calls, checks Google Calendar availability and books appointments end to end through **tool-calling workflows**.
- **Designed multi-step tool-calling workflows** with custom endpoints, webhook handlers, timezone-aware scheduling and defensive parsing of nested model/tool payloads.
- Built an LLM-assisted data pipeline processing **~20,000 business records** using self-hosted Firecrawl and **Gemini 2.5 Flash** for scraping, extraction, summarisation and structured enrichment.
- **Diagnosed and resolved production failures** including Playwright browser-pool exhaustion, PostgreSQL query ambiguity and a Content-Security-Policy issue blocking WebRTC.
- **Owned deployment and infrastructure** across Hetzner Cloud, Docker, Coolify, PostgreSQL and Vercel, including DNS and production configuration.

Advanced RAG Portfolio Chatbot | In Development

LangChain, LangGraph, Runnables, Hybrid Retrieval, Multimodal RAG

- Building the retrieval workflow with **LangChain Runnables and LangGraph**, using modular ingestion, retrieval and generation steps that can be composed, traced and extended independently.
- Implemented **hybrid search** combining dense semantic retrieval with lexical search, followed by **reranking** to prioritize the most relevant context before generation.
- Integrated a **basic multimodal RAG** path that indexes and retrieves relevant **images alongside text**, preserving source metadata so retrieved evidence can be traced back to the originating document/page.

Additional Software Projects | Next.js, TypeScript, Node.js

- Canva-style design editor: canvas editing, authentication, AI image generation and Stripe payments.
- Personal finance tracker: dashboards and type-safe APIs with Next.js, Hono.js and React Query.

CLIENT EXPERIENCE

Freelance Video Editor & Colorist | Fiverr Profile | 2020 - Present

- Built a strong Fiverr track record with a **5.0 rating across 100+ reviews, 130 completed orders and 72 clients**, earning Fiverr's Choice for consistent delivery quality.
- Worked directly with independent international clients outside Fiverr, delivering **500+ video exports** while managing requirements, revisions and deadlines; later shifted focus to **AI engineering** through hands-on work with LLMs, RAG and automation systems.

EDUCATION & CERTIFICATIONS

Bachelor of Computer Applications (BCA) | Indira Gandhi National Open University, Delhi | 2019 - 2021

TensorFlow Developer Certificate - Google, 2023 | Diploma in 3D & VFX - Image Institute, 2021